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Comparative Study of Solar Radiation Measurements in Tucuman, Argentina, With Reanalyses and Satellite-Based Datasets

Reanalysis and satellite-based datasets are easily accessible sources for evaluating global solar radiation at the surface, particularly useful in regions with scarce observational data. This study evaluates their performance considering two reanalyses (ERA5 and MERRA-V2) and one satellite-based (Copernicus Atmosphere Monitoring Service (CAMS)) datasets, in comparison with ground-based observations at Tucuman, Argentina, over the period 2006–2023. This area has very limited measurements, and those that exist are not readily accessible. The statistical metrics such as mean relative absolute error (MRAE), mean relative error (MRE), and squared correlation coefficient (r^2) were assessed for hourly, daily, and monthly values. Results indicate that CAMS provides the closest match to measurements, followed by ERA5, which in turn outperforms MERRA-V2. All datasets exhibit a tendency to overestimate measured radiation values. Temporal variability analysis reveals a good agreement between datasets, particularly in capturing diurnal and seasonal cycles. Less accuracy is obtained in the interannual variation, which is expected considering that it is relatively weak and of climatic origin, linked in particular with cloud cover. This study contributes to the validation of reanalysis and satellite-based datasets in the southern hemisphere. Enhanced validation efforts and adjustments based on ground measurements are necessary to further refine these datasets for practical applications in solar energy optimization in the region here analyzed where measured data are limited.

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1 Introduction

Solar radiation is the primary energy source on Earth, driving physical, chemical, and biological processes. It plays a key role in climate modeling and simulation, and knowing its time and space variability is essential for the efficient planning and operation of solar energy systems [1]. Surface-based measurements, such as those from pyranometers, offer the most accurate estimation of solar radiation. However, their spatial coverage is often sparse and uneven, leaving many regions of the world with limited data [2,3].

Several methods have been developed to address the lack of ground-based data, but challenges related to data availability and accuracy persist. In addition, installing and maintaining pyranometers in general requires significant time and resources. However, it is essential to have a precise estimation of solar energy potential to fulfill energy demands and for the correct dimensioning of devices to optimize energy use [4]. Ruiz Arias et al. [5] found that a spacing of 100–150 km between ground stations can give unbiased gridded estimates. However, Zerenka et al. [6] suggest using satellite data if the closest station is over 20–30 km away when precise time and location data are needed.

To overcome these issues, global gridded datasets derived from satellite remote sensing and reanalysis products have become essential tools. Satellite observations provide high spatial resolution (less than 10 km) and relatively accurate estimates of surface solar radiation. On the other hand, reanalysis models, which assimilate diverse data sources, offer broad spatial coverage and long temporal records, making them particularly useful for solar radiation estimation in data-scarce regions (measurements more than 100 km apart) [7]. However, the heterogeneity of data sources and differences in

assimilation schemes can introduce biases and uncertainties into reanalysis datasets, needing validation to characterize and quantify these deviations [8,9].

While there are several papers evaluating reanalysis and satellite-based solar irradiance datasets for northern hemisphere regions [8–11], there are fewer for the southern hemisphere, and in particular for Argentina in southern South America, where the solar radiation ground measurement data available are scarce (more than 100 km apart) [12–14]. Maps of global average daily solar radiation using interpolation techniques based on ground-based measurement data have been first generated for Argentina using all the information available from the Argentinean Solarimetric Network, pyranometers, and measured sunshine hours by the Servicio Meteorológico Nacional (SMN) [15–18]. However, these solar radiation products depend on the interpolation techniques used and on the quantity and distribution of pyranometers, which are not fully available in Argentina due to high installation and maintenance costs [16]. Next, comparative analyses of measurements with reanalysis and satellite data started to appear. Ramirez Camargo and Dorner [12] evaluate solar radiation in the Lerma Valley, Salta, Argentina, by comparing pyranometer measurements in the region with satellite-derived (by remote sensing of reflectivity) and reanalysis data. Their results highlight that satellite data show the closest agreement with ground measurements, demonstrating their reliability for mapping global solar radiation in the region. Sarmiento et al. [14] focus on the province of Salta, Argentina, and develop a geographic information system tool for solar irradiation using data from satellite sources, empirical models, and ground stations. Over a 7-year period, comparisons showed that satellite data offer high reliability for characterizing solar resources. Ceballos

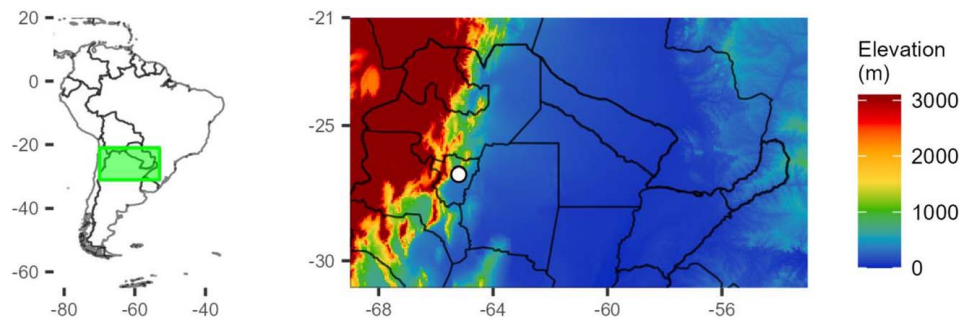


Fig. 1 (Left panel) South America with the region containing Tucumán highlighted in green (indicated with a square). (Right panel) Topographic map of northern Argentina indicating the location of Tucumán (white dot).

et al. [19] tested the GL1.2 Model that runs at the Brazilian National Institute for Space Research (CPTEC/INPE) for characterizing the solar radiation regime over Pampa Region in Argentina in the period 2011–2018, obtaining a very good agreement. The GL1.2 model is a satellite-based physical algorithm developed and operated by CPTEC/INPE to estimate surface solar irradiance over South America. It uses visible channel imagery from GOES-East satellites (GOES-12, GOES-13, GOES-16) at 0.04 deg (~ 4 km) resolution. The review work by Bragagnolo et al. [20] explores the deployment of solar energy in Argentina, emphasizing the gap between its great solar potential and the limited adoption of solar technologies. It highlights the reliance on satellite and reanalysis databases for solar resource estimation due to insufficient ground-based data. The study points out the need for better integration of these datasets into the planning and implementation of solar energy projects to bridge Argentina's solar energy deployment gap.

This study provides a comparative analysis of surface solar radiation measured in Tucumán, South America (26.8°S , 65.1°W , Fig. 1), from 2006 to 2023 with two reanalyses and a satellite-based dataset, as a further contribution to the validation of these data sources. Reanalyses are ERA5 (European Centre for Medium-Range Weather Forecasts (ECMWF) Reanalysis v5) [20], and MERRA-V2 (Modern-Era Retrospective Analysis for Research and Applications, Version 2) [21], and the satellite-derived solar radiation is from Copernicus Atmosphere Monitoring Service (CAMS) [22,23]. Argentina, extending from 22°S and 55°S in latitude and having a significant west-to-east elevation gradient (as can be noticed in the map of Fig. 1), exhibits considerable variability in its solar resource. Thus, our study is an additional contribution to the analysis of this resource in our region and its representation by satellite and reanalysis databases.

This article is organized as follows: Sec. 2 describes the data used; Sec. 3 presents the methodology for the comparative analysis; Sec. 4 reports the results; followed by the discussion and conclusions in Sec. 5.

2 Data

2.1 Data Measured at Tucumán. The global total radiation data (R_s (W/m^2)) used in this study were provided by the Estacion Experimental Agroindustrial Obispo Colombres (EEAOC), an autonomous research and development organization under the Ministry of Productive Development of the government of Tucumán dedicated to supporting agricultural and agro-industrial sectors. Data consist of surface measurements obtained using a photovoltaic solarimeter with silicon photodiode sensors [17], sensitive in the spectral band of 300–1100 nm but is calibrated for 300–3000 nm solar radiation.

The sensors are integrated into the Davis Vantage Pro-2 automatic weather station installed at the central agrometeorological observatory located in El Colmenar (26.8°S , 65.2°W), Las Talitas, Tucumán Province, Argentina, at 451 m above sea level. Their resolution is $1 \text{ W}/\text{m}^2$, and the accuracy is 5% at full scale.

In addition, regular cleaning and operational checks of the sensors are performed to ensure the accuracy and reliability of the collected data. The values are published online.² Figure 1 shows Tucumán location in the map of South America and the North of Argentina. The period of data availability is January 2006 to December 2023.

The station records data every 15 min, which amounts to a total of 631,104 data considering four values per hour of each day of the period considered. The missing data amount to 4608 in total, that is, $\sim 0.7\%$. A quality control following Long and Shi [24] was performed eliminating data exceeding the “extremely rare limits,” which amounts to 1697 data ($\sim 0.3\%$ of the total). Hourly averages were estimated then without applying a minimum data requirement. That is, the hourly value was calculated even with one data in the four values per hour interval.

Daily total radiation, in MJ/m^2 , was calculated as the sum of the 24 h values. Days with more than 2 missing hours in the interval 8–18 local time (LT) (interval with R_s values greater than zero) were discarded, which amounts to 35 days. So from the 6574 total days in the period, we are left with 6539 values. The missing data are distributed across the months, generally with 1–3 missing values per month except for March 2022, which has 11 missing values. Monthly mean radiation, in MJ/m^2 , was obtained as the average of the daily total values.

2.2 Data From ERA5 Reanalysis. The ERA5 data used in this study correspond to surface solar radiation downward (ssrd) for the period 2006–2023 [25]. This reanalysis dataset provides hourly estimates of ssrd with a latitude–longitude resolution of 0.25° . From the four ERA5 grid points surrounding the EEAOC station, an inverse distance weighting (IDW) interpolation was performed to obtain the corresponding time series at Tucumán's location. The offset between universal time coordinated (UTC) and LT for Argentina was taken into account, considering that the solar local time is 1-h shifted from the UTC-3 time zone.

The ssrd represents the shortwave solar radiation reaching a horizontal surface at the Earth's surface and refers to the sum of direct and diffuse solar radiation in the plane. It is considered a good approximation of the modeled value of radiation that would be measured by a pyranometer, but integrating the spectral band of 300–4000 nm. The additional radiation in the longer near infrared range (3000–4000 nm), not included in the pyranometer measured values, can contribute about 1% of the total solar energy reaching the Earth's surface, which is practically negligible. Additional details on this are presented in Sec. 5.

2.3 Data From MERRA-V2 Reanalysis. The MERRA-V2 data used correspond to surface shortwave net radiation flux (SWGNT) for the period 2006–2023 [21]. This reanalysis dataset provides hourly estimates of SWGNT with a latitude–longitude

²<https://agromet.eaac.gov.ar/>.

resolution of $0.5 \text{ deg} \times 0.625 \text{ deg}$. From the four MERRA-V2 grid points surrounding the EEAOC station, an IDW interpolation was performed to obtain the corresponding time series at the station location. The offset between UTC and LT for Argentina was taken into account, considering that the solar local time is 1 h shifted from the UTC-3 time zone.

The SWGNT represents the shortwave solar radiation reaching a horizontal surface at the Earth's surface and is also considered a good approximation of the value measured by a pyranometer. It integrates the 300–4000 nm spectral band, as in the case of ERA5.

2.4 Satellite-Based Data From CAMS. Global horizontal irradiance (GHI) from the CAMS satellite-based model was retrieved from the official SoDaPro site [22,23]. In this case, the time series has a resolution of 15 min and is obtained for the exact location of the EEAOC station.

GHI refers to the all-sky global irradiation, that is, the surface solar downward irradiation integrated over the 285–2800 nm spectral band on a horizontal surface [22,23]. So, in this case, the spectral band that matches with the measured data is very close.

3 Methodology

Measured R_s values were compared to ERA5, MERRA-V2, and CAMS considering hourly, daily, and monthly means, using three comparative statistic parameters [26–28]: mean relative absolute error (MRAE), mean relative error (MRE), and squared correlation coefficient (r^2), defined by

$$\text{MRAE}(\%) = 100 \frac{1}{n} \sum \frac{|y - x|}{x} \quad (1)$$

$$\text{MRE}(\%) = 100 \frac{1}{n} \sum \frac{(y - x)}{x} \quad (2)$$

$$r^2 = \frac{[n(\sum xy) - (\sum x)(\sum y)]^2}{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]} \quad (3)$$

where “ x ” corresponds to the measured values, “ y ” to the reanalysis or satellite-based values, and “ n ” to the number of data points in the time series. MRAE is a measure of the mean deviation of reanalysis

or satellite data from R_s in percentage. The optimal value is 0, indicating that both series are identical. MRE measures the average bias of reanalysis or satellite datasets overestimating or underestimating measured R_s depending on its sign: positive or negative, respectively, in percentage. Its optimal value is 0, but in this case, it does not indicate that both series are identical since there can be a compensation of errors. r^2 measures the explained variance in R_s by the reanalysis or satellite data time variation. So, it measures how similar the variability in one series is to that of another. It is bounded by 1 and 0, where 1 is the optimal value indicating that both time series vary identically.

These statistics were assessed considering hourly mean, daily total, and monthly mean data along the period 2006–2023. These assessments were repeated in the monthly case considering each month separately along 2006–2023 in order to analyze the interannual variation only.

The standard deviation (σ) of daily means within each month was also evaluated using these comparative statistics. This is an important parameter since it measures the variability of solar energy availability. Thus, σ provides insight into the range of solar irradiance values experienced over the month, which is important for designing and sizing solar energy systems linked to storage requirements, for example.

4 Results

4.1 Descriptive Statistics. To provide a visual overview of the analyzed data, Fig. 2 shows the monthly mean time series of R_s , together with ERA5, MERRA-V2, and CAMS values, and σ of the monthly values. In all cases, σ presents a similar seasonal variation in phase with R_s , being maximum during summer months and minimum during winter. In relation to the seasonality, in addition to the fact that the lower standard deviation is generally expected when a variable has lower values, in this case, there is also the effect of cloudiness. Over 1 month, the solar declination does not vary much, and consequently, the daily solar radiation incident at the top of the atmosphere at a given location is expected to remain almost constant. Thus, the day-to-day variation of the global radiation received on the Earth's surface during a given month should be primarily due to the variation of cloudiness, and to a much lesser extent, due to other factors such as the variation of the atmospheric

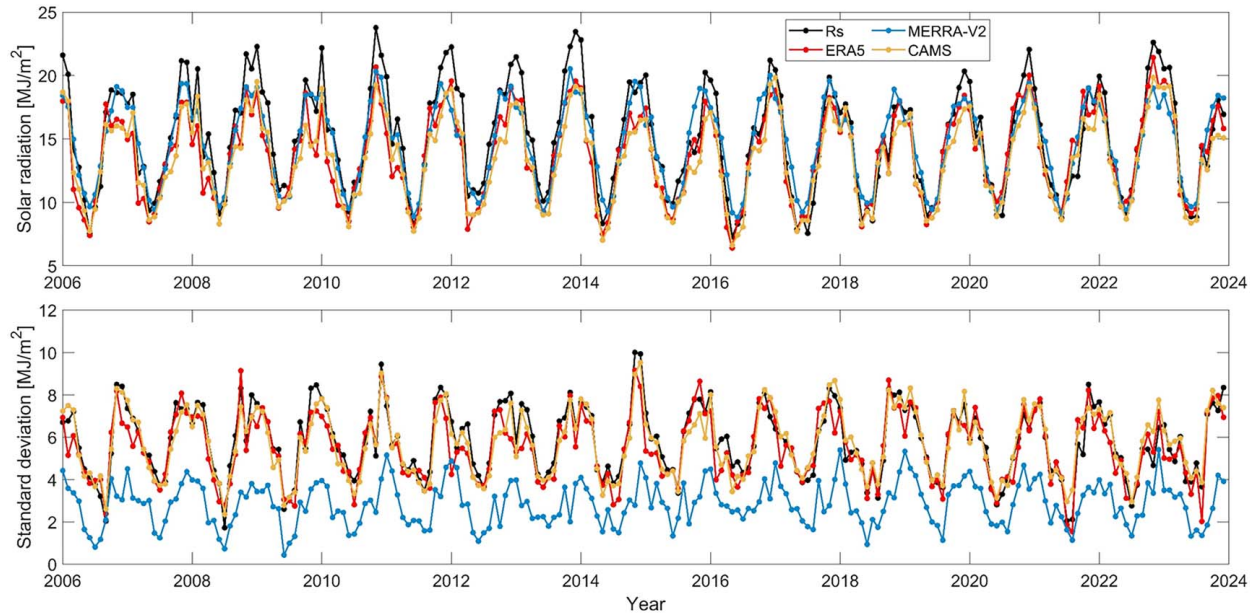


Fig. 2 Monthly mean (upper panel) and standard deviation (lower panel), both in MJ/m^2 , along 2006–2023 of the daily total surface global solar radiation values obtained from the pyranometer R_s (black), ERA5 (red/medium gray), MERRA-V2 (blue/dark gray), and CAMS (orange/light gray)

Table 1 Comparative statistic values of Rs with ERA5, MERRA-V2, and CMAS considering hourly mean (*H*), daily (*D*), and monthly mean (*M*) values

	ERA5			MERRA			CMAS		
	<i>H</i>	<i>D</i>	<i>M</i>	<i>H</i>	<i>D</i>	<i>M</i>	<i>H</i>	<i>D</i>	<i>M</i>
MRE (%)	46.7	28.1	15.9	85.4	61.3	23.6	9.4	20.1	11.0
MRAE (%)	65.9	34.6	16.4	102.2	65.9	23.6	58.0	23.9	11.2
r^2	0.80	0.70	0.82	0.79	0.52	0.91	0.88	0.86	0.94

Note: MRAE, mean relative absolute error (%), MRE, mean relative error (%), and r^2 , squared correlation coefficient.

water vapor and dust [29]. This is consistent with the fact that Tucuman, located in the northwest region of Argentina, has a monsoon precipitation regime with a greater concentration of rainfall in the summer months and dry months during the colder season [27,30,31]. This would imply cloudier days in summer than in winter.

4.2 Comparative Analysis of Mean Values. Table 1 lists the values of MRAE, MRE and r^2 between Rs and the reanalyses and satellite datasets, for the hourly mean, daily total, and monthly mean cases. Clearly, all databases overestimate Rs. The best agreement is shown by CAMS and the weakest by MERRA-V2. Figure 3 shows the dispersion between Rs and each database. To better reveal the overestimation or underestimation of the reanalysis and satellite data, the first bisector is plotted in each graph. This figure highlights the overestimation already evident from MRE values.

Regarding temporal variability, the variability of Rs hourly, daily, and monthly values is well explained in all the cases with r^2 greater than 0.8 (0.79 in the case of MERRA-V2). This is reasonable considering that most of their time variability is explained just by the diurnal and seasonal cycles. Here again, the best agreement is shown by CAMS.

The comparative statistics based on monthly means were estimated also considering separate time series for each month. Their values are shown in Fig. 4. Since MRAE and MRE are almost identical due to the predominance of overestimation in monthly values, only MRE results are presented. It is notorious, in particular, the marked decrease in r^2 centered in the month of August.

This can be explained by the fact that the seasonal behavior, which enhances correlations when series with concatenated months is considered, is absent from the time series for each month separately. Evidently the variation of each monthly mean from year to year, i.e., their interannual variation, is not well represented in all the months. August stands out as the month with the worst agreement in the interannual variation. It is also noteworthy that during this month, ERA5 shows a weaker agreement with Rs than MERRA-V2 according to the MRE values. In fact, the interannual variation from year to year is expected to be exclusively due to atmospheric conditions and cloud cover variability, which do not seem to be captured correctly by the reanalyses and satellite databases. Nevertheless, the interannual variability of the monthly average for a given month is very low, as will be seen later in the analysis of the standard deviation.

4.3 Comparative Analysis of Standard Deviation Values.

Table 2 lists the values of MRAE, MRE, and r^2 between σ of the daily means of Rs within each month along the period 2006–2023, and σ of the reanalysis and satellite data. Unlike the monthly average, where all the datasets overestimate Rs, σ is underestimated by all of them, especially MERRA-V2, which showed a clear underestimation already in Fig. 2. In the case of CAMS, the underestimation is less than 1%. The temporal variability is determined mainly by seasonality, so high values of r^2 are expected except, again, in the case of MERRA-V2.

The interannual variability of Rs monthly means is also valuable for the long-term planning of solar energy systems. Thus, σ of the monthly means along the whole period considered for each

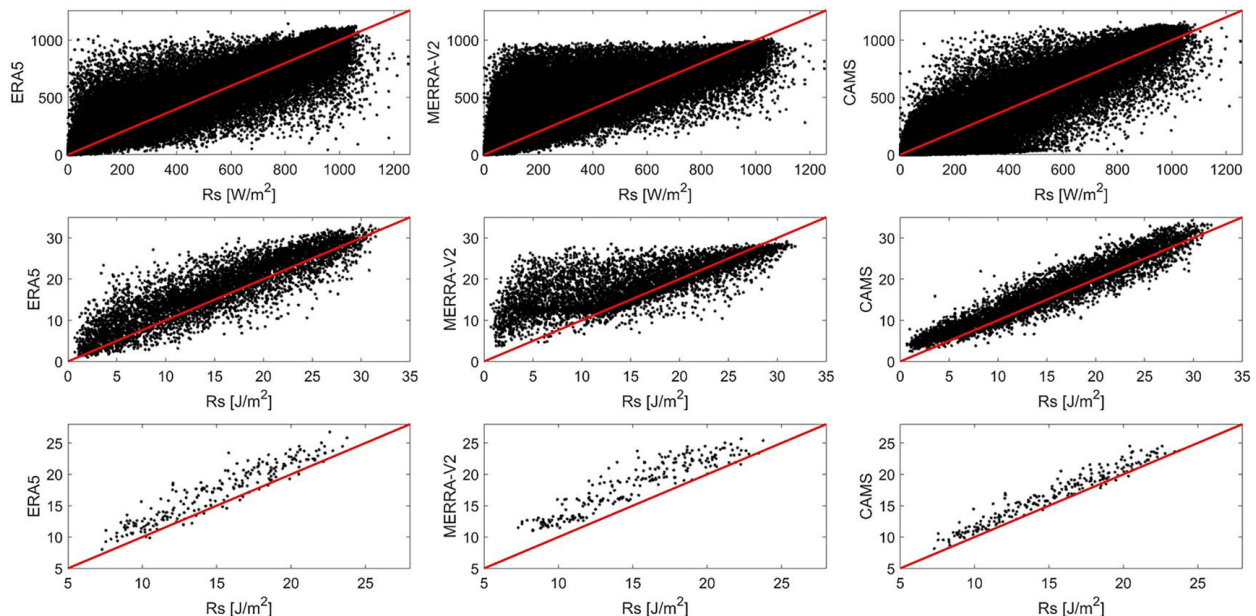


Fig. 3 Dispersion diagrams for hourly (upper panels), daily (middle panels), and monthly (lower panels) values, of Rs with ERA5 (first column panels), MERRA-V2 (second column panels), and CAMS (third column panels). The red line (straight lines) corresponds to the first bisector meaning perfect match.

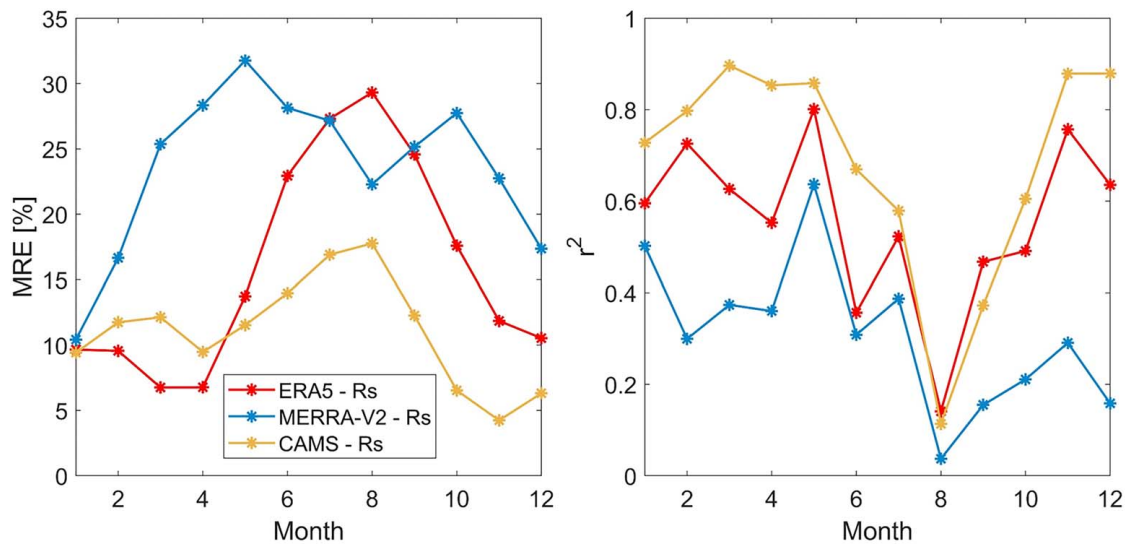


Fig. 4 Mean relative error, MRE (%) (left panel), and square correlation coefficient, r^2 (right panel), between Rs and ERA5 (red/medium gray), MERRA-V2 (blue/dark gray), and CAMS (orange/light gray) for each month considering monthly mean values the period 2006–2023

Table 2 Comparative statistic values of monthly standard deviations (σ) of Rs with ERA5, MERRA-V2, and CMAS obtained from daily mean values within a month

	ERA5	MERRA-V2	CAMS
MRE (%)	−4.4	−50.8	−0.5
MRAE (%)	10.2	50.8	8.1
r^2	0.84	0.54	0.88

Note: MRAE, mean relative absolute error (%), MRE, mean relative error (%), and r^2 , squared correlation coefficient.

month was estimated and plotted in Fig. 5, together with the average of Rs monthly means. We focus in this case on the relative value of σ . When comparing a given month along different years, a low σ value is expected since on average its variation should be determined by the interannual variation of the solar radiation at the top of the atmosphere, which is extremely weak even along the quasi-

decadal cycle of solar activity [32]. This is noticed from comparing σ values in Fig. 2 (which correspond to the daily variability within a month) to those in Fig. 5 (which correspond to the annual variability for each month). The σ values in the latter case may be attributed mainly to climatic effects linked to cloud cover variability from year to year.

5 Discussion and Conclusions

This study evaluates the performance of reanalysis and satellite-based datasets (ERA5, MERRA-V2, and CAMS) in estimating surface solar radiation at Tucuman, Argentina, in South America, by comparing them with ground-based observations over an 18-year period (2006–2023). The results provide insights into the strengths and limitations of these data sources.

CAMS data are the most accurate among the analyzed datasets, showing lower relative errors at all time resolutions and the highest correlations with the observed data. When only the

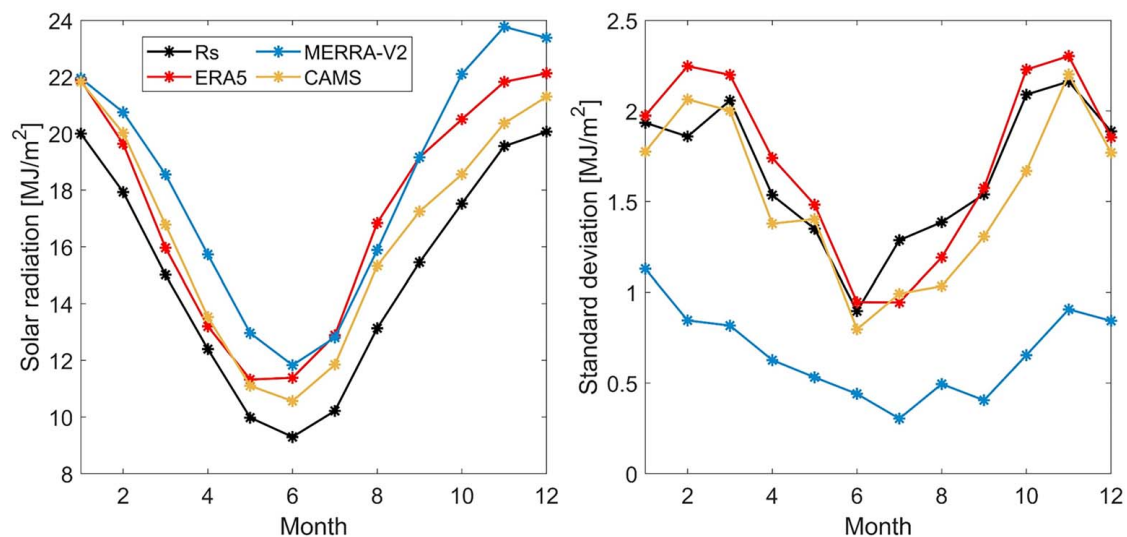


Fig. 5 Average along 2006–2023 of monthly mean (left panel) and standard deviation (right panel) for each month of Rs (black), ERA5 (red/medium gray), MERRA-V2 (blue/dark gray), CAMS (orange/light gray)

interannual variation is analyzed, this agreement decreases, especially in August.

In the case of reanalyses, ERA5 performs better than MERRA-V2. Its relatively finer spatial resolution may have contributed to its higher accuracy compared to MERRA-V2. Seasonal variation is well captured; however, systematic overestimation is a common issue in both reanalysis datasets. One might consider the spectral mismatch between the data, which cover the range 300–3000 nm, and the reanalysis databases, which cover 300–4000 nm. Calculating the integral of the spectrum over each range, by numerically integrating Planck's law or the American Society for Testing and Materials (ASTM) G-173 spectra, which represent the terrestrial solar spectral irradiance at the surface under a set of specified atmospheric conditions and air mass 1.5, yields a ratio of measured to reanalyses integral of ~ 0.99 , in both cases. This indicates a difference of about 1%, which is practically negligible. In fact, applying these factors to the respective dataset values before the statistical comparisons improves MRAE and MRE by only $\sim 1\%$.

Given its stronger alignment with observed data, CAMS is recommended as the most reliable source for solar radiation studies in the region for estimating monthly means. For hourly and daily values, CAMS is less accurate but provides the best fit. ERA5 is also a valuable tool for long-term climate and energy planning. The relatively high errors and lower correlation observed in the case of MERRA-V2 indicate that it is less appropriate for Tucuman. In addition, it underestimates σ , especially in short-term (inter-monthly) variability, likely influenced by local factors such as cloud cover.

Regarding the variation from year to year in a given month, the performance of each dataset varies greatly by month, being particularly poor in August, with r^2 values around 0.1 or less. However, this variability is weak compared to seasonal or daily variability.

Similar analyses conducted in other regions of the world show error results comparable to those obtained here, particularly regarding the overestimation by reanalyses databases [33] and the better agreement of ERA5 data [34]. There are numerous other studies [3,9,10,35,36], most of them focused on stations in the northern hemisphere. Among the fewer studies focused on the Southern Hemisphere, in addition to those mentioned in the Introduction section, it is that by Mabasa et al. [37] who analyzed 13 radiometric stations in South Africa to validate global solar irradiance from three satellite-based datasets, one of which is CAMS, as well as the two reanalyses used in this work. Similar to this study, they demonstrated that CAMS performs best, while MERRA-V2 was the overall worst-performing dataset.

In a global study by Yang et al. [38], 690 stations worldwide were analyzed in comparison to seven representative datasets including ERA5 and MERRA-V2 and satellite databases, considering monthly mean values. Overall, these databases tended to overestimate R_s , with MERRA-V2 being the least accurate. One notable finding from this study was the decreasing long-term trend they observed during the period 2001–2010, though they concluded that their databases might not be suitable for long-term analysis. Yang and Bright [39] also analyzed worldwide stations covering a period of 27 years and presented a validation study on eight-gridded irradiance products consisting of six satellite-based datasets, including CAMS, and the two reanalyses used in the present study. As in this work, they concluded that satellite-derived products are better than reanalysis products.

In agreement with other similar studies carried out in Argentina [12,14,18,20], we consider that it is essential to improve the Argentinean national solar radiation database. However, there are other available sources of information derived from satellites and reanalyses that can contribute to filling the current gaps with varying accuracies, which can serve different purposes. The findings in this study contribute to the validation of satellite and reanalysis datasets in the southern hemisphere and serve as a foundation for future research on improving solar radiation models. Enhanced validation efforts and adjustments based on ground measurements are

necessary to further refine these datasets for practical applications in solar energy optimization.

In the future, this comparative analysis will be expanded to better characterize the error associated with using reanalyses and satellite datasets, not only by extending the analysis period and increasing the number of stations but also by taking into account the varying degrees of cloud cover, following the approach proposed by Sianturi et al. [34], along with the correction factors suggested by Tahir et al. [33], to enable the incorporation of reanalysis and satellite-based values with greater reliability into the study of solar radiation in the region.

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Funding Data

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Conflict of Interest

There are no conflicts of interest.

Data Availability Statement

The observational solar radiation data can be obtained upon request from J. D. Forciniti (jorgeforciniti@eeaac.org.ar). Solar radiation data in Argentina, though limited, are freely available; however, they are not compiled in a single open-access repository. Institutions including the SMN, the EEAOC, and some research groups manage these datasets. Data are provided upon request or shared through publications, reports, and irradiance maps, rather than as raw downloadable datasets. Despite this decentralized system, they can be obtained by contacting the respective institutions.

ERA5 data: Copernicus Climate Change Service, Climate Data Store, (2023): ERA5 hourly data on single levels from 1940 to present. Copernicus Climate Change Service (C3S) Climate Data Store (CDS), 10.24381/cds.adbb2d47, <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels?tab=overview>, Accessed June 11, 2025.

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ASTM G-173 reference solar spectral irradiance data is provided by the US Department of Energy (DOE)/National Renewable Energy Laboratory (NREL)/ALLIANCE, <https://www.nrel.gov/grid/solar-resource/spectra-am1.5>

Nomenclature

- x = measured values
- y = estimated values from reanalysis or satellite-based values

N = number of data points in the time series
 r^2 = squared correlation coefficient
 R_s = measured global total radiation measured: hourly (W/m^2) and daily (MJ/m^2) values
 S_{srd} = surface solar radiation downwards from ERA5: hourly (W/m^2) and daily (MJ/m^2) values
 GHI = global horizontal irradiance from CAMS: hourly (W/m^2) and daily (MJ/m^2) values
 $MRAE$ = mean relative absolute error (%)
 MRE = mean relative error (%)
 $SWGNT$ = surface shortwave net radiation flux from MERRA-V2: hourly (W/m^2) and daily (MJ/m^2) values

Greek Symbol

σ = standard deviation

Abbreviations

ASTM = American Society for Testing and Materials
 ASTM G-173 = American Society for Testing and Materials G-173 spectra provided as standard tables for reference solar spectral irradiance at air mass 1.5, direct normal and hemispherical for a 37 deg tilted surface
 CAMS = Copernicus Atmosphere Monitoring Service
 CPTEC = Centro de Previsão de Tempo e Estudos Climáticos (Center for Weather Forecasting and Climate Studies), at INPE, Brazil
 EEAO = Estación Experimental Agroindustrial Obispo Colombres (Obispo Colombres Agricultural Experimental Station), at Tucuman, Argentina
 ERA5 = European Centre for Medium-Range Weather Forecasts (ECMWF) Reanalysis v5
 GL1.2 = satellite-based model developed and operated by CPTEC/INPE
 INPE = Instituto Nacional de Pesquisas Espaciais (National Institute for Space Research), in Brazil
 LT = local time
 MERRA-V2 = Modern-Era Retrospective Analysis for Research and Applications, Version 2
 SMN = Servicio Meteorológico Nacional (National Meteorological Service), in Argentina
 UTC = universal time coordinated

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